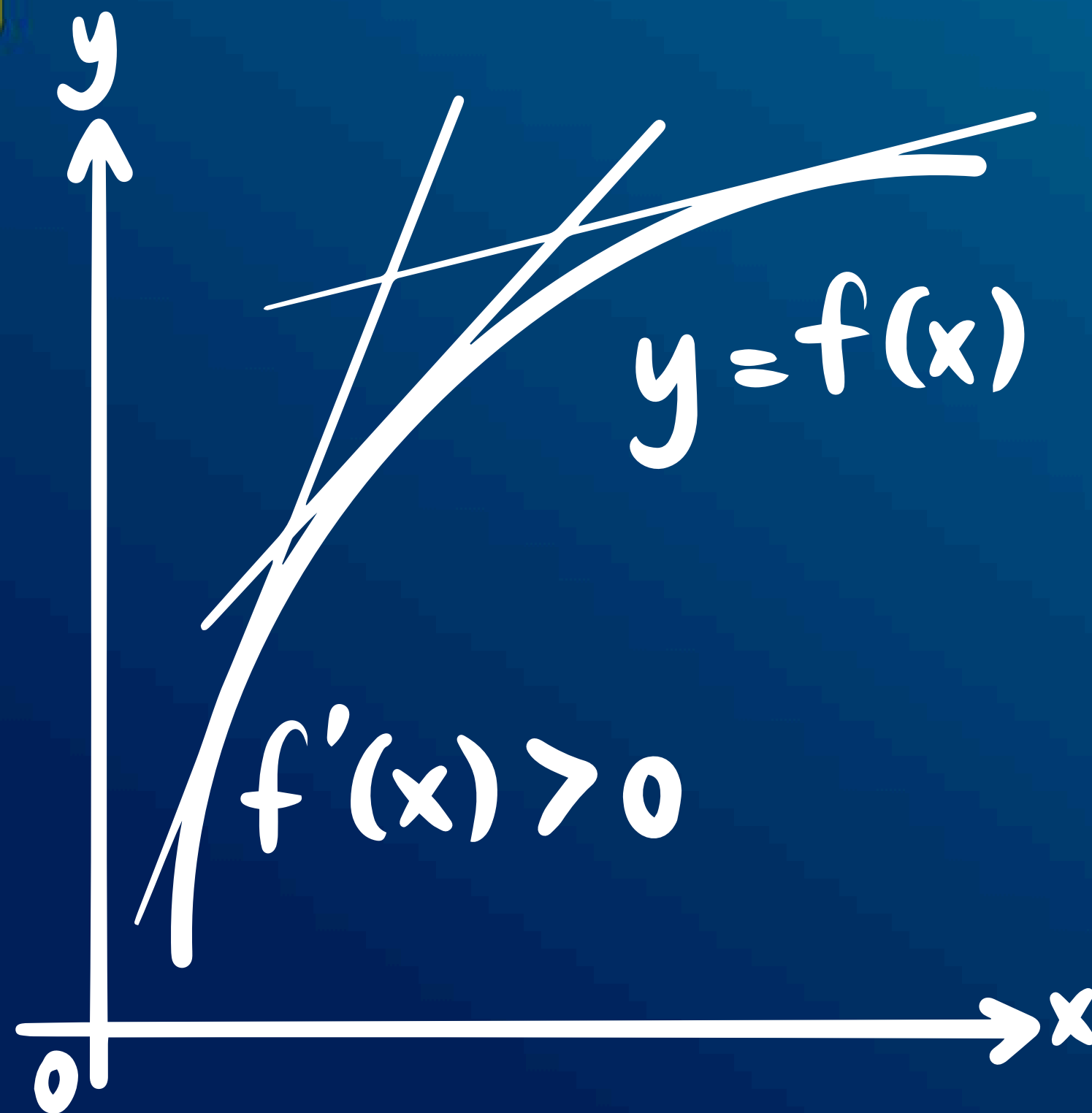
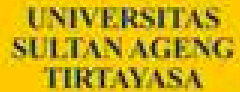




Calculus 1

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[illegible]

Theorem A Main Limit Theorem

Let n be a positive integer, k be a constant, and f and g be functions that have limits at c . Then

1. $\lim_{x \rightarrow c} k = k$;
2. $\lim_{x \rightarrow c} x = c$;
3. $\lim_{x \rightarrow c} kf(x) = k \lim_{x \rightarrow c} f(x)$;
4. $\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$;
5. $\lim_{x \rightarrow c} [f(x) - g(x)] = \lim_{x \rightarrow c} f(x) - \lim_{x \rightarrow c} g(x)$;
6. $\lim_{x \rightarrow c} [f(x) \cdot g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$;
7. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided $\lim_{x \rightarrow c} g(x) \neq 0$;
8. $\lim_{x \rightarrow c} [f(x)]^n = \left[\lim_{x \rightarrow c} f(x) \right]^n$;
9. $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow c} f(x)}$, provided $\lim_{x \rightarrow c} f(x) > 0$ when n is even.

EXAMPLE 1 Find $\lim_{x \rightarrow 3} 2x^4$.

$$\lim_{x \rightarrow 3} 2x^4 = 2 \lim_{x \rightarrow 3} x^4 = 2 \left[\lim_{x \rightarrow 3} x \right]^4 = 2[3]^4 = 162$$

EXAMPLE 2 Find $\lim_{x \rightarrow 4} (3x^2 - 2x)$.

SOLUTION

$$\begin{aligned} \lim_{x \rightarrow 4} (3x^2 - 2x) &= \lim_{x \rightarrow 4} 3x^2 - \lim_{x \rightarrow 4} 2x = 3 \lim_{x \rightarrow 4} x^2 - 2 \lim_{x \rightarrow 4} x \\ &= 3 \left(\lim_{x \rightarrow 4} x \right)^2 - 2 \lim_{x \rightarrow 4} x = 3(4)^2 - 2(4) = 40 \end{aligned}$$

Limit

Limit Theorems

EXAMPLE 3

Find $\lim_{x \rightarrow 4} \frac{\sqrt{x^2 + 9}}{x}$.

SOLUTION

$$\begin{aligned}
 \lim_{x \rightarrow 4} \frac{\sqrt{x^2 + 9}}{x} &\stackrel{(7)}{=} \frac{\lim_{x \rightarrow 4} \sqrt{x^2 + 9}}{\lim_{x \rightarrow 4} x} \stackrel{(9,2)}{=} \frac{\sqrt{\lim_{x \rightarrow 4} (x^2 + 9)}}{4} \stackrel{(4)}{=} \frac{1}{4} \sqrt{\lim_{x \rightarrow 4} x^2 + \lim_{x \rightarrow 4}} \\
 &\stackrel{(8,1)}{=} \frac{1}{4} \sqrt{\left[\lim_{x \rightarrow 4} x \right]^2 + 9} \stackrel{(2)}{=} \frac{1}{4} \sqrt{4^2 + 9} = \frac{5}{4}
 \end{aligned}$$

EXAMPLE 4

If $\lim_{x \rightarrow 3} f(x) = 4$ and $\lim_{x \rightarrow 3} g(x) = 8$, find

$$\lim_{x \rightarrow 3} [f^2(x) \cdot \sqrt[3]{g(x)}]$$

SOLUTION

$$\begin{aligned}
 \lim_{x \rightarrow 3} [f^2(x) \cdot \sqrt[3]{g(x)}] &\stackrel{(6)}{=} \lim_{x \rightarrow 3} f^2(x) \cdot \lim_{x \rightarrow 3} \sqrt[3]{g(x)} \\
 &\stackrel{(8,9)}{=} \left[\lim_{x \rightarrow 3} f(x) \right]^2 \cdot \sqrt[3]{\lim_{x \rightarrow 3} g(x)} \\
 &= [4]^2 \cdot \sqrt[3]{8} = 32
 \end{aligned}$$

Limit

Limit Theorems

Theorem B Substitution Theorem

If f is a polynomial function or a rational function, then

$$\lim_{x \rightarrow c} f(x) = f(c)$$

provided $f(c)$ is defined. In the case of a rational function, this means that the value of the denominator at c is not zero.

EXAMPLE 5

Find $\lim_{x \rightarrow 2} \frac{7x^5 - 10x^4 - 13x + 6}{3x^2 - 6x - 8}$.

SOLUTION

$$\lim_{x \rightarrow 2} \frac{7x^5 - 10x^4 - 13x + 6}{3x^2 - 6x - 8} = \frac{7(2)^5 - 10(2)^4 - 13(2) + 6}{3(2)^2 - 6(2) - 8} = -\frac{11}{2}$$

EXAMPLE 6

Find $\lim_{x \rightarrow 1} \frac{x^3 + 3x + 7}{x^2 - 2x + 1}$

SOLUTION

$$\lim_{x \rightarrow 1} \frac{x^3 + 3x + 7}{x^2 - 2x + 1} = \lim_{x \rightarrow 1} \frac{x^3 + 3x + 7}{(x - 1)^2} \quad \text{limit is } +\infty$$

Limit

Limit Theorems

Theorem C

If $f(x) = g(x)$ for all x in an open interval containing the number c , except possibly at the number c itself, and if $\lim_{x \rightarrow c} g(x)$ exists, then $\lim_{x \rightarrow c} f(x)$ exists and $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x)$.

EXAMPLE 7

Find $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$.

SOLUTION

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} = \lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(\sqrt{x} + 1)}{\sqrt{x} - 1} = \lim_{x \rightarrow 1} (\sqrt{x} + 1) = \sqrt{1} + 1 = 2$$

EXAMPLE 8

Find $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x^2 + x - 6}$

SOLUTION

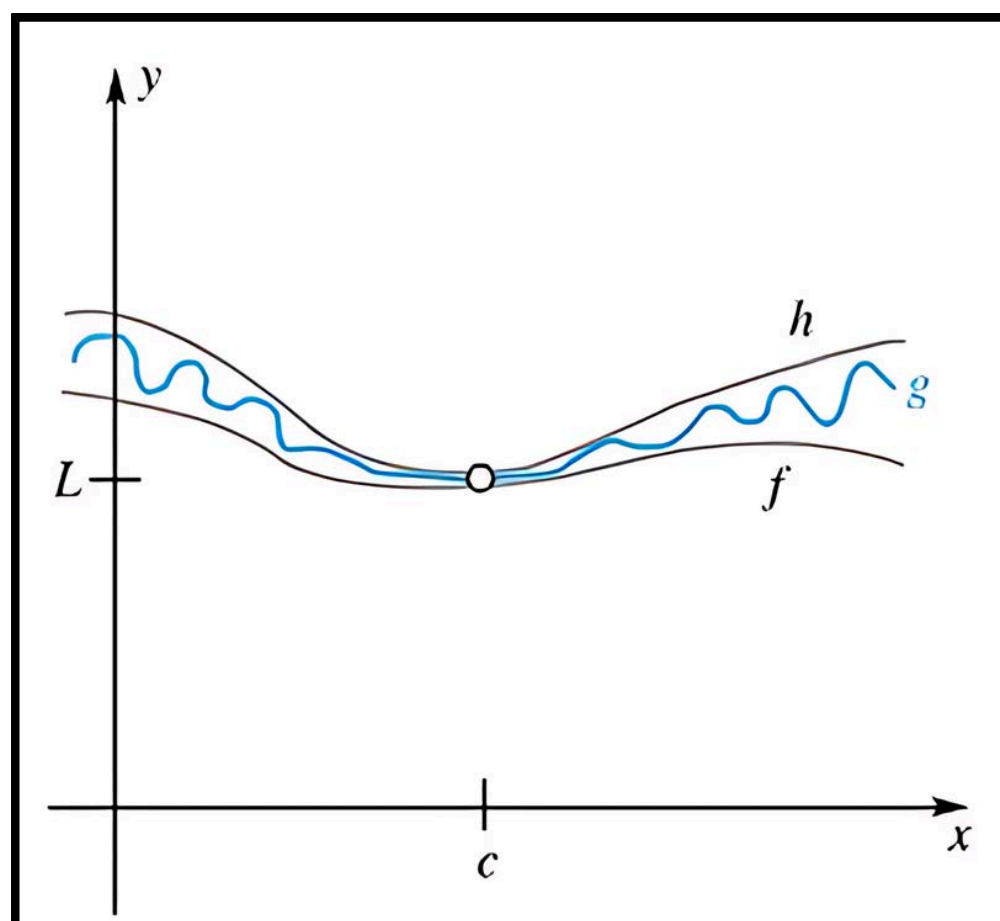
$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x^2 + x - 6} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 5)}{(x - 2)(x + 3)} \\ &= \lim_{x \rightarrow 2} \frac{x + 5}{x + 3} = \frac{7}{5} \end{aligned}$$

Limit

Limit Theorems

Theorem D Squeeze Theorem

Let f , g , and h be functions satisfying $f(x) \leq g(x) \leq h(x)$ for all x near c , except possibly at c . If $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} g(x) = L$.



EXAMPLE 9 Assume that we have proved $1 - x^2/6 \leq (\sin x)/x \leq 1$ for all x near but different from 0. What can we conclude about $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?

SOLUTION Let $f(x) = 1 - x^2/6$, $g(x) = (\sin x)/x$, and $h(x) = 1$. It follows that $\lim_{x \rightarrow 0} f(x) = 1 = \lim_{x \rightarrow 0} h(x)$ and so, by Theorem D,

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Limit

Limit Theorems (Time to participate)

$$\lim_{x \rightarrow -3} \frac{4x^3 + 1}{7 - 2x^2}$$

1

$$\lim_{y \rightarrow 2} \left(\frac{4y^3 + 8y}{y + 4} \right)^{1/3}$$

2

$$\lim_{u \rightarrow -2} \frac{u^2 - ux + 2u - 2x}{u^2 - u - 6}$$

3

$$\lim_{x \rightarrow \pi} \frac{2x^2 - 6x\pi + 4\pi^2}{x^2 - \pi^2}$$

4

$$\lim_{x \rightarrow -1} \frac{x^3 - 6x^2 + 11x - 6}{x^3 + 4x^2 - 19x + 14}$$

5

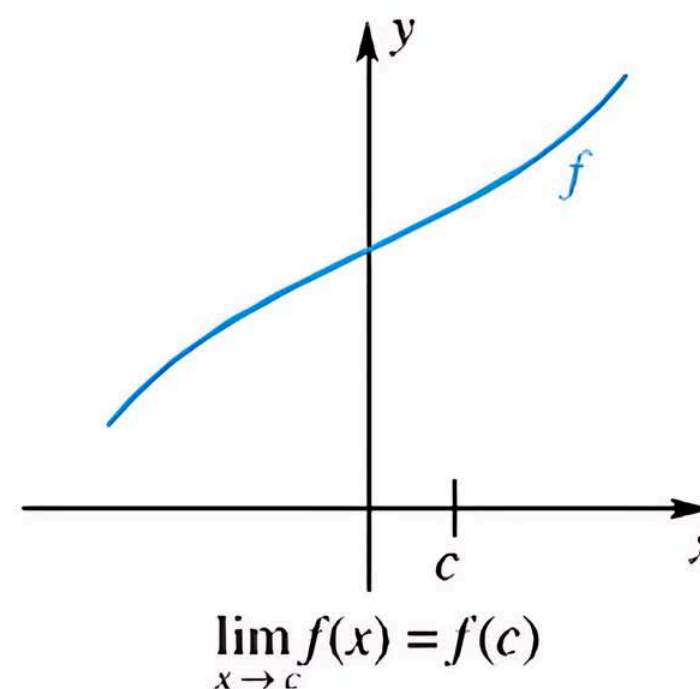
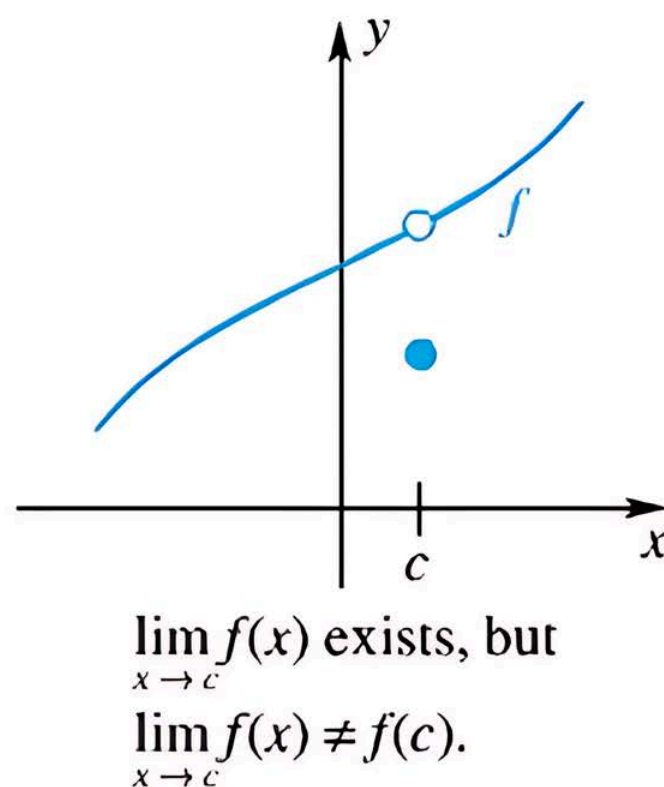
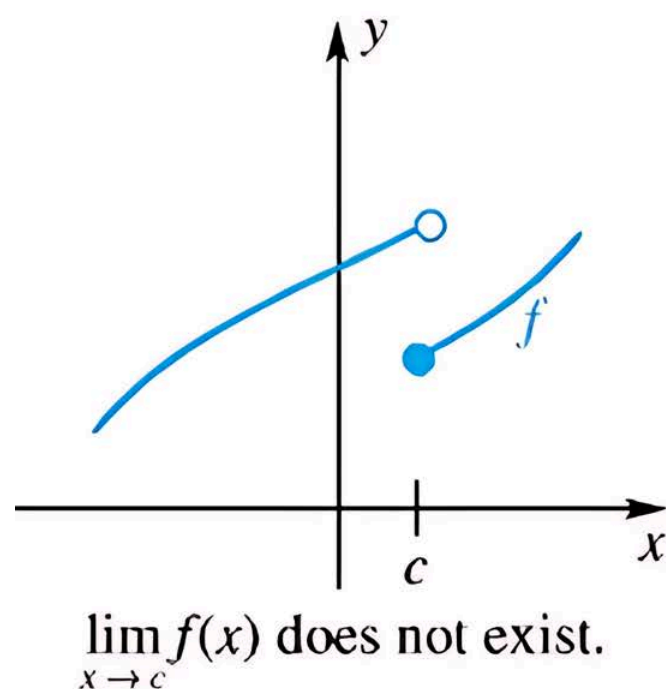
Continuity

Continuity Function

Definition Continuity at a Point

Let f be defined on an open interval containing c . We say that f is **continuous** at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$



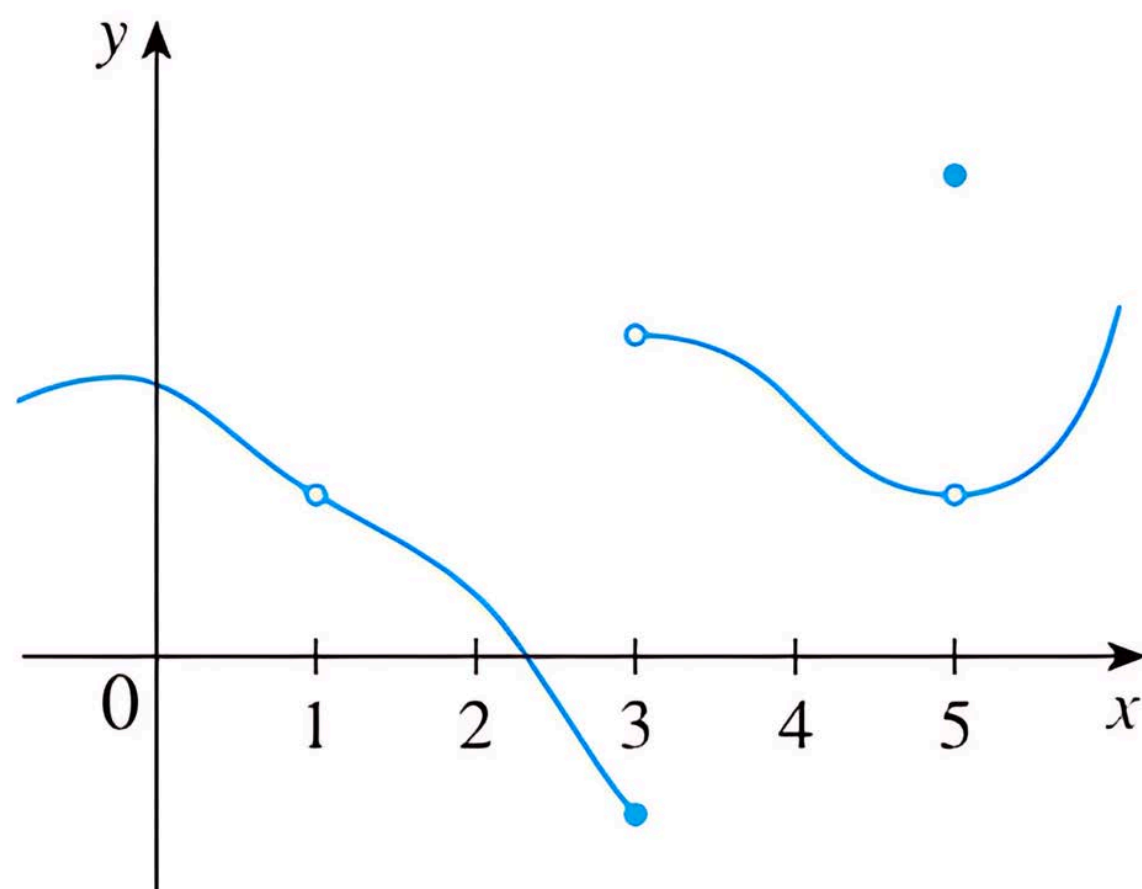
We mean by this definition to require three things:

1. $\lim_{x \rightarrow c} f(x)$ exists,
2. $f(c)$ exists (i.e., c is in the domain of f), and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Continuity

Continuity Function

EXAMPLE 1 Figure 2 shows the graph of a function f . At which numbers is f discontinuous? Why?



SOLUTION It looks as if there is a discontinuity when $a = 1$ because the graph has a break there. The official reason that f is discontinuous at 1 is that $f(1)$ is not defined.

The graph also has a break when $a = 3$, but the reason for the discontinuity is different. Here, $f(3)$ is defined, but $\lim_{x \rightarrow 3} f(x)$ does not exist (because the left and right limits are different). So f is discontinuous at 3.

What about $a = 5$? Here, $f(5)$ is defined and $\lim_{x \rightarrow 5} f(x)$ exists (because the left and right limits are the same). But

$$\lim_{x \rightarrow 5} f(x) \neq f(5)$$

So f is discontinuous at 5.

FIGURE 2

Continuity

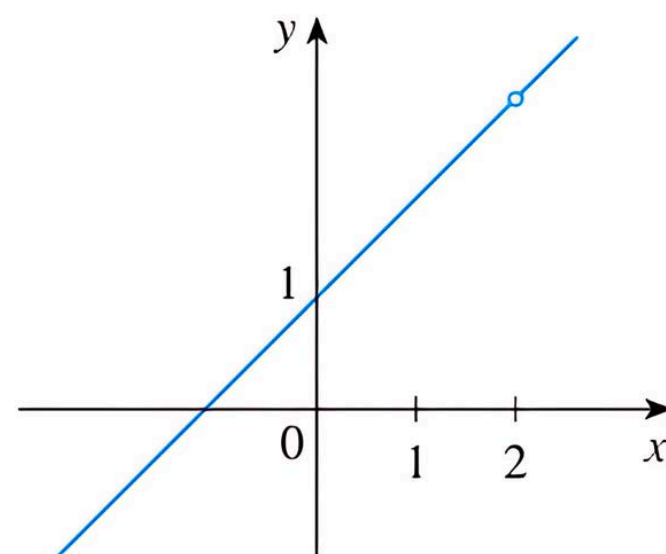
Continuity Function

EXAMPLE 2 Where are each of the following functions discontinuous?

$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

SOLUTION

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2. Later we'll see why f is continuous at all other numbers.



$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

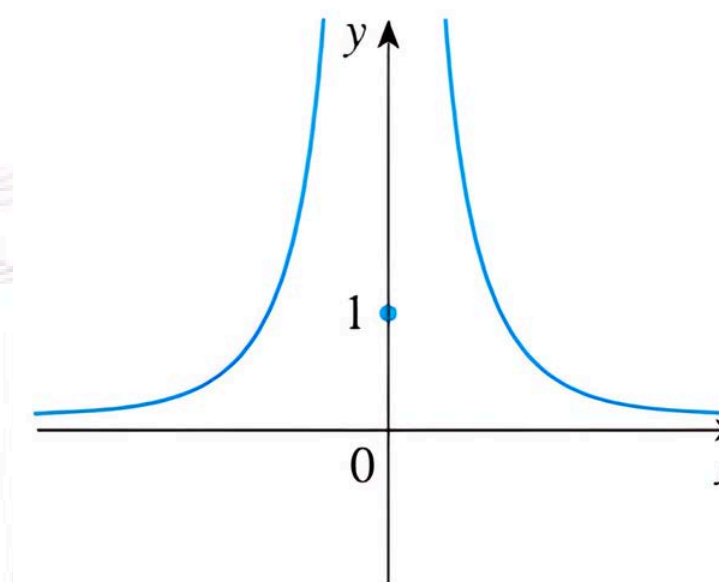
SOLUTION

(b) Here $f(0) = 1$ is defined but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2}$$

does not exist.

So f is discontinuous at 0.



$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

SOLUTION

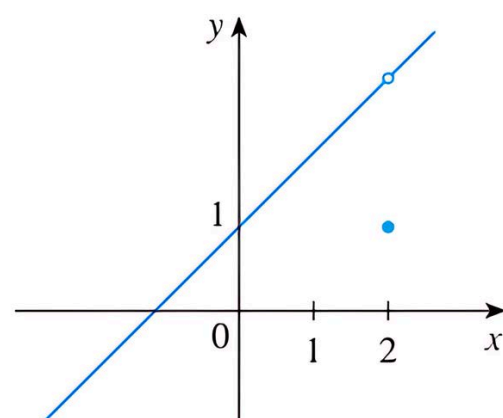
(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

exists. But

$$\lim_{x \rightarrow 2} f(x) \neq f(2)$$

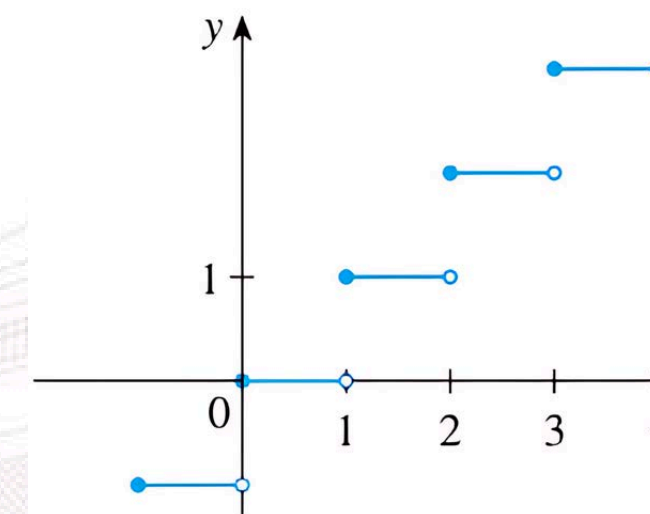
so f is not continuous at 2.



$$(d) f(x) = \llbracket x \rrbracket$$

SOLUTION

(d) The greatest integer function $f(x) = \llbracket x \rrbracket$ has discontinuities at all of the integers because $\lim_{x \rightarrow n} \llbracket x \rrbracket$ does not exist if n is an integer.





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Continuity

Continuity Function

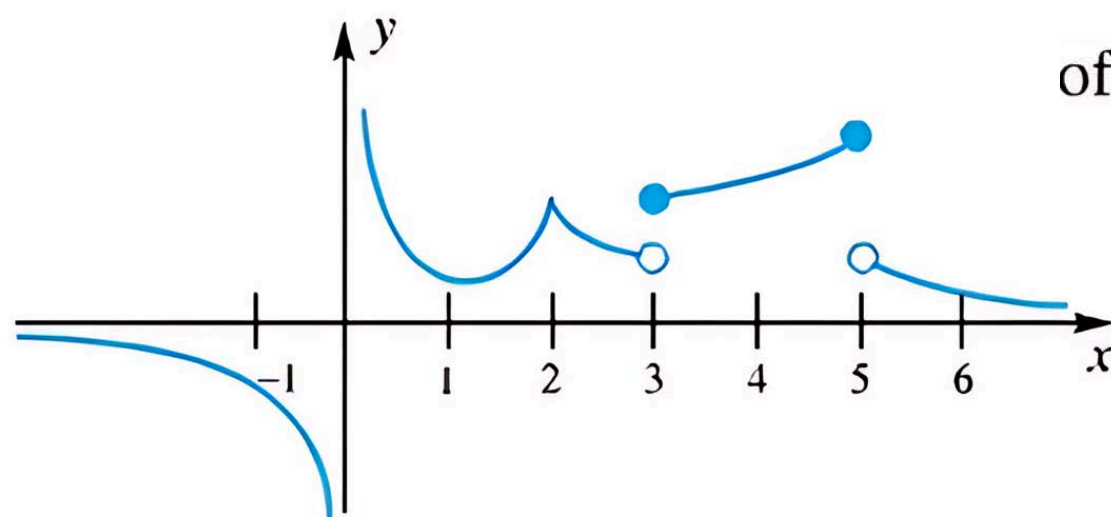
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Definition Continuity on an Interval

The function f is **right continuous** at a if $\lim_{x \rightarrow a^+} f(x) = f(a)$ and **left continuous** at b if $\lim_{x \rightarrow b^-} f(x) = f(b)$.

We say f is **continuous on an open interval** if it is continuous at each point of that interval. It is **continuous on the closed interval** $[a, b]$ if it is continuous on (a, b) , right continuous at a , and left continuous at b .

EXAMPLE 1 Using the definition above, describe the continuity properties of the function whose graph is sketched in Figure



SOLUTION The function appears to be continuous on the open intervals $(-\infty, 0)$, $(0, 3)$, and $(5, \infty)$, and also on the closed interval $[3, 5]$.

Continuity

Continuity Function

EXAMPLE 2 What is the largest interval over which the function defined by $g(x) = \sqrt{4 - x^2}$ is continuous?

SOLUTION The domain of g is the interval $[-2, 2]$. If c is in the open interval $(-2, 2)$, then g is continuous at c by Theorem E; hence, g is continuous on $(-2, 2)$. The one-sided limits are

$$\lim_{x \rightarrow -2^+} \sqrt{4 - x^2} = \sqrt{4 - \left(\lim_{x \rightarrow -2^+} x\right)^2} = \sqrt{4 - 4} = 0 = g(-2)$$

and

$$\lim_{x \rightarrow 2^-} \sqrt{4 - x^2} = \sqrt{4 - \left(\lim_{x \rightarrow 2^-} x\right)^2} = \sqrt{4 - 4} = 0 = g(2)$$

This implies that g is right continuous at -2 and left continuous at 2 . Thus, g is continuous on its domain, the closed interval $[-2, 2]$. ■



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Continuity

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EXAMPLE 3 Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ interval $[-1, 1]$ is continuous on the interval $[-1, 1]$.

SOLUTION If $-1 < a < 1$, then using the Limit Laws, we have

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) \\ &= 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} \\ &= 1 - \sqrt{\lim_{x \rightarrow a} (1 - x^2)} \\ &= 1 - \sqrt{1 - a^2} \\ &= f(a)\end{aligned}$$

Thus, by Definition f is continuous at a if $-1 < a < 1$. Similar calculations show that

$$\lim_{x \rightarrow -1^+} f(x) = 1 = f(-1) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = 1 = f(1)$$

so f is continuous from the right at -1 and continuous from the left at 1 . Therefore, according to Definition f is continuous on $[-1, 1]$.

Continuity

Continuity Function (Time to participate)

state whether the indicated function is continuous at 3. If it is not continuous, tell why.

$$r(t) = \begin{cases} \frac{t^3 - 27}{t - 3} & \text{if } t \neq 3 \\ 27 & \text{if } t = 3 \end{cases}$$

$$r(t) = \begin{cases} \frac{t^3 - 27}{t - 3} & \text{if } t \neq 3 \\ 23 & \text{if } t = 3 \end{cases}$$

$$f(t) = \begin{cases} t^2 - 9 & \text{if } t \leq 3 \\ (3 - t)^2 & \text{if } t > 3 \end{cases}$$

what points, if any, are the functions discontinuous?

$$f(x) = \begin{cases} x & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x \leq 1 \\ 2 - x & \text{if } x > 1 \end{cases}$$

$$g(x) = \begin{cases} x^2 & \text{if } x < 0 \\ -x & \text{if } 0 \leq x \leq 1 \\ x & \text{if } x > 1 \end{cases}$$

1

2

3

4

5



SEE YOU NEXT WEEK !

